Teacher Wu Kuan | Real Exam #1 — Taiwan GSAT Math A (2026) Q6: Isosceles, but Check Collinearity

English lecture notes · YouTube Short q9IBNMTv35A

1. The trap question

A(2, −2), B(−1, 2). How many points C on y = −6 make △ABC isosceles?

2. The method

• Three cases: CA = AB, CB = AB, CA = CB.
• AB = 5. CA = 5 → C(5, −6), (−1, −6). CB = 5 → none (B is 8 from the line). CA = CB → perpendicular bisector → C(−15/2, −6).
• Final check: (5, −6) lies on line AB → not a triangle. Answer: 2.

3. Parent check method

Ask your child to check every point: “Do these three points actually form a triangle?”

Practice

1. A(0, 0), B(6, 0). Which point C on y = 4 has CA = CB?
2. A(1, 3), B(4, 7). Which points C on the x-axis have CA = AB?
3. A(1, 2), B(3, 6), C(k, 10) are collinear. Find k.
4. A(0, 0), B(3, 4). How many points C on y = 8 make △ABC isosceles?
5. A(0, 0), B(4, 0). How many points C on y = x make △ABC isosceles?

Answers

1. (3, 4)
2. (5, 0) and (−3, 0)
3. k = 5
4. 2
5. 4

Saber Math · Teacher Wu Kuan (YouTube: Saber Math) · Free first-week trial of Kuan Weekly practice (email sign-up): https://kuan-weekly.vercel.app/sample · More: https://kuanmath.vercel.app · SAT notes: DM IG @saber_math